

MICROTEC Colloidal Silica Sol Range

Aqueous dispersions of amorphous SiO₂ nanoparticles · 3 to 30 nm particle size · 10 to 40% SiO₂ content · Alkaline, Neutral & Acidic grades

ABOUT MICROTEC

Microtec is Sterling Chemicals' proprietary range of colloidal silica sols — stable aqueous dispersions of ultra-fine amorphous SiO₂ nanoparticles. The range spans multiple particle sizes, silica concentrations, and surface chemistries to serve a broad spectrum of industrial applications.

Manufactured at our dedicated facility in Ankleshwar, Gujarat — one of India's largest chemical manufacturing zones — Microtec grades are produced to consistent specifications and exported to customers globally.

WHY STERLING

- **Direct Manufacturer:** No intermediary. Buy directly from the producer with full access to technical and production teams.
- **Tailormade Grades:** Particle size, silica content, pH, and stabiliser chemistry customised to your formulation requirements.
- **Consciously Priced:** Performance equivalent to global MNC suppliers at significantly more competitive pricing.
- **Proven Track Record:** Trusted OEM supplier to leading specialty chemical companies and distributors across 10+ countries.

MANUFACTURING CAPABILITIES

STABILISING ION	pH CHARACTER	PARTICLE SIZE	SiO ₂ CONTENT
— Sodium — Potassium — Aluminium — Ammonium	— Acidic (pH 2 – 5) — Neutral (pH 5 – 7) — Alkaline (pH 8 – 11)	3 – 30 nm Tailored to application requirements. Finer particles for higher surface area; coarser for controlled gelation.	10 – 40% Supplied at customer's required concentration. Higher solids reduce freight cost per unit of SiO₂.

KEY APPLICATIONS

■ Paper & Board

Retention and drainage microparticle for paper machine wet-end chemistry. Works alongside PAM and cationic starch to improve first-pass retention and drainage speed.

■ Waterproofing & Construction

Concrete densification and surface hardening. Reacts with Ca(OH)₂ to form additional C-S-H gel, reducing porosity and increasing surface durability.

■ Coatings & Films

Additive for water-based paints and functional coatings. Improves scratch and abrasion resistance, barrier properties, and coating hardness at low addition levels.

■ CRGO Silicon Steel

Insulation coating binder for cold-rolled grain-oriented (CRGO) electrical steel. Provides the dielectric insulation layer between laminations in transformer cores.

■ Refractory & Investment Casting

Binder for ceramic shell moulds, refractory castables, and ceramic fibre products. Acidic grades (Microtec-40AC) available for alkali-sensitive systems.